INCH-POUND
MIL-C-82762
21 November 1989

MILITARY SPECIFICATION

CUTTER, DELAY, PROPELLANT ACTUATED, MLU-58/B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the Propellant Actuated Cutter MLU-58/B (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For
MIL-W-5625	Webbing, Textile, Nylon, Tubular
MIL-S-13572	Spring, Helical, Compression and Extension
MIL-C-13739	Composition, Delay
MIL-P-20444	Primer, Percussion, M42 Parts for and Loading, Assembling and Packing
MIL-P-22264	Powder, Ignition, Gasless, A-1A
MIL-P-25732	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Limited Service at 275 °F
MIL-W-83420	Wire, Rope, Flexible, for Aircraft Control

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-1168	Lot Numbering of Ammunition

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

DL6261278	Cutter, Propellant Actuated, 4 Second Delay, MLU-58/B

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CAGE Code 14083)

8873332	Cutter tester

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5000.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR-43	Preparation of Quality Assurance Provisions

(Unless otherwise indicated, copies of WR's are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM A108	Steel Bars, Carbon, Cold Finished Standard Quality
ASTM A276	Stainless and Heat-Resisting Steel Bars and Shapes
ASTM B211	Aluminum Alloy Bars, Rods and Wire
ASTM B221	Aluminum Alloy Extruded Bars, Rods, Wire, Shapes, and Tube

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The cutters shall comply with all requirements specified on drawings listed on DL6261278 and with all requirements specified in applicable specifications (see 4.8.1).

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.6, 6.2 and 6.4).

3.3 Materials. All materials used in the manufacture of the cutters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure has been obtained from the cognizant Navy design activity prior to manufacture. When alternate methods of manufacture are specified in the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Primary components. For the purpose of this specification, the delay assembly Drawing 6261282 is defined as a primary component. Only primary components from a single lot shall be used in a production lot of cutters. One primary component production lot may be used in more than one cutter production lot.

3.5 Primer.

3.5.1 Primer x-ray. All primers shall be x-rayed and screened prior to assembly into the primer retainer (see 4.8.4).

3.5.2 Primer sensitivity. The primer shall fire when drop tested at an energy level of 26 inch-ounces in a primer retainer assembly (see 4.8.5).

3.6 Spring load and spring embrittlement. The springs shall not break and shall meet the requirements of Drawing 8875988 when tested as specified in 4.8.6 and 4.8.7.

3.7 Corrosion resistance. All component parts (except for the delay body, Drawing 6261279; washer (retainer), Drawing 6261280; and primer retainer, Drawing 8875985) shall not exhibit white corrosion products after one hundred hours of salt spray testing. The delay body, washer, and primer retainer shall not exhibit white corrosion products after 24 hours of salt spray testing (see 4.8.8).

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3.8 Leakage.

3.8.1 Airtightness. All knife delay assemblies shall be leak tested and shall not exhibit a leakage rate in excess of 1×10^{-4} cc of air per second at a pressure differential of one atmosphere (see 4.8.9).

3.8.2 Waterproofness. The knife delay element shall not show any evidence of leakage when tested as specified in 4.8.10.

3.9 Lanyard assembly. The lanyard assembly shall withstand a tensile load of 200 pounds minimum without separation of parts (see 4.8.11).

3.10 Radiographic examination. After completion of assembly, all cutters shall be radiographically examined for proper assembly and presence of components. Any evidence of misassembly or missing parts shall be cause for rejection of that assembly (see 4.8.12).

3.11 Staking. The staking between the firing pin assembly and case shall withstand a torque of 25 inch-pounds without turning or loosening of the components (see 4.8.13).

3.12 Vibration. The cutter shall withstand the vibration test without evidence of any disintegration of the primer, delay column, propellant, or damage of any component (see 4.8.14).

3.13 Withdrawal of cotter pin. The safety cotter pin shall not withdraw from the cutter when a pull force of 10 pounds minimum is applied to the pin. The pin shall be withdrawn when a pull force of up to 30 pounds maximum is applied to the pin (see 4.8.15).

3.14 Functional.

3.14.1 Initiation. The firing pin shall not be released when a force of 10 pounds is gradually applied to the lanyard assembly. The cutter shall function when the firing pin is released. The firing pin shall release when a force equivalent to dropping a 20 pound weight from a height of 3 ± 1/3 inches is applied to the lanyard assembly (see 4.8.16).

3.14.2 Cutting. The cutter shall completely sever two reefing lines when functioned (see 4.8.17).

3.14.3 Delay Time. The delay time shall be as indicated in table I when tested in accordance with 4.8.17 (see 6.9.2).

TABLE I. *Delay time.*

Temperature (°F)	Delay time (seconds)	
	Minimum	Maximum
−65	2.44	6.91
+70	2.77	5.33
+165	2.34	5.00

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3.14.4 Functioning over temperature range. The cutter shall function satisfactorily without failure (see 6.9.1) throughout the temperature range of −65 to +165 °F when tested in accordance with 4.8.17.

3.14.5 Functioning at high setback. The cutter shall function satisfactorily, as evidenced by complete severing of the reefing lines, when a shock force of 700 g minimum is applied to the cutter (see 4.8.18).

3.14.6 Misfire. There shall be no misfire.

3.15 Workmanship. Cutters shall be free of cracks, splits, incomplete or damaged threads, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.15.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, incomplete or damaged threads, porosity or any similar defect.

3.15.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.15.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

3.16 Packaging, packing, and marking. The packaging and packing of cutters and the container marking shall be as specified in section 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein unless disapproved by the Government. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspection. Inspection shall be in accordance with MIL-A-2550 and MIL-STD-453.

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4.2 Classification of inspections. Inspection shall be in accordance with MIL-A-2550, except as specified herein. Inspections specified herein shall be classified as follows:

a. Material inspection (see 4.3)
b. First article inspection (see 4.6)
c. Quality conformance inspection (see 4.7)

4.3 Material inspection. Material inspection shall consist of certification supported by verifying data that the materials listed in table II, used in fabrication of the cutters, are in accordance with the applicable specification or requirement prior to such fabrication (see 6.3).

TABLE II. *Material inspection.*

Material	Requirement paragraph	Applicable specification
Aluminum	3.3	ASTM B211 ASTM B221
Ignition composition	3.3	MIL-P-22264
O-ring	3.3	MIL-P-25732
Spring steel wire	3.3	MIL-S-13572
Steel	3.3	ASTM A108 ASTM A276
Delay composition	3.3	MIL-C-13739
Primer	3.3	MIL-P-20444
Steel cable	3.3	MIL-W-83420

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-810.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major, or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and Major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified as minor.

4.6 First article inspection. First article inspection shall be performed after award of contract and prior to production. After inspection and provisional acceptance at the source, sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract. First article inspection shall be performed on sample units which have been manufactured in the same manner using the same materials, equipment, processes, and procedures as to be used in the regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply (see 6.2 and 6.4).

4.6.1 Sample size. Unless otherwise specified, the first article sample shall consist of 28 assembled units, 10 complete sets of components and subassemblies (i.e. 10 each of every component and 10 each of the lanyard and firing pin assemblies and 10 knife delay assemblies), and one set of metal components without coating or protective finish for dimensional and visual examination and tests.

4.6.2 Inspection routine. The first article sample shall be subjected to the inspections specified in table III.

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TABLE III. *First article inspection.*

Inspection	Requirement paragraph	Test method paragraph	Number of samples to be inspected
Visual and dimensional inspection of components	3.1	4.8.1	10 components
Hardness	3.3	4.8.2	5 components
Spring load	3.6	4.8.6	5
Embrittlement of spring	3.6	4.8.7	5
Corrosion resistance	3.7	4.8.8	3 components
Airtightness	3.8.1	4.8.9	10
Waterproofness of knife delay assembly	3.8.2	4.8.10	10
Lanyard assembly pull test	3.9	4.8.11	10
Radiographic	3.10	4.8.12	28
Staking	3.11	4.8.13	28
Vibration	3.12	4.8.14	12
Non-withdrawal and withdrawal of cotter pin	3.13	4.8.15	21
Firing pin release and non-release	3.14.1	4.8.16	28
Functioning over temperature range	3.14.4	4.8.17	28
Functioning at high setback	3.14.5	4.8.18	7

4.6.3 First article sample failure. The first article sample shall meet all the requirements in table III. The Government reserves the right to terminate its inspection upon any failure of any assembly or component in the sample to comply with the stated requirements. In the event of failure of any sample, the Government reserves the right to require the contractor to take corrective action and submit a new sample or portion thereof.

4.7 Quality conformance inspection.

4.7.1 Inspection of product. Inspection of product shall consist of Group A inspection and Group D inspection. Group A and Group D inspections shall be done prior to delivery.

4.7.1.1 Submission of product. A lot shall consist of an identifiable group of assembled cutters containing metal parts assemblies produced by a homogeneous process and loaded with one lot of knife delay assemblies. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.7.1.2 Lot size. A lot shall consist of an identifiable group of assembled cutters of not more than 1200 units.

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4.7.1.3 Group A inspection. Group A inspection shall consist of the inspections specified in table IV in the order shown (see 6.9.4).

4.7.1.3.1 Examination. Examination for critical defects shall be done on a sample size of 200. For lots of 200 or under, 100 percent inspections for all critical defects shall be done. Examination for major or minor defects shall be performed on a defect basis (see 6.9.3) in accordance with the classification on each applicable drawing of DL6261153 or as contained herein, using the sampling plan and associated criteria specified in the contract (see 6.5). In addition to the characteristics shown on assembly drawing 6261278, examination for characteristics listed in table V shall also be made. All non-conforming material shall be rejected.

TABLE IV. *Group A inspection.*

Inspection	Requirement paragraph	Test method paragraph	Quantity tested	Accept/reject criteria
Visual and dimensional examination	3.1	4.8.1	Sample per 6.4	4.7.1.3.2
Protective finish	3.3	4.8.3	Per Spec	Per Spec
Primer x-ray	3.5.1	4.8.4	All	No failures
Primer sensitivity	3.5.2	4.8.5	See 4.7.1.3.4	No failures
Spring load	3.6	4.8.6	10	No failures
Embrittlement of springs	3.6	4.8.7	10	No failures
Airtightness	3.8.1	4.8.9	10	4.7.1.3.2
Waterproofness of knife assembly	3.8.2	4.8.10	65	4.7.1.3.2
Lanyard assembly pull test	3.9	4.8.11	All	4.7.1.3.2
Radiographic inspection of assembled cutters	3.10	4.8.12	All	4.8.12
Withdrawal and non-withdrawal of cotter pin	3.13	4.8.15	25	4.7.1.3.2
Release and non-release of firing pin	3.14.1	4.8.16	All	4.7.1.3.2

TABLE V. *Classification of assembly defects.*

Classification	Defect	Method of inspection
Critical		
1	All parts are present and assembled correctly	Visual/x-ray
Major		
101	Assembly damaged	Visual
102	Stake inadequate	Visual
103	Marking clear and legible	Visual
Minor		
None defined		

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4.7.1.3.2 Component or assembly failures. Any components or assembled cutters failing to meet requirements shall be cause for rejection of that item. In the waterproofness of knife delay assembly test, if two failures occur, the lot is rejected. If only one failure is found, a second sample of 130 assemblies shall be tested. When a total of two defectives is encountered in the combined first and second sample, testing will be stopped and the lot shall be rejected. In the non-withdrawal and withdrawal of safety cotter pin test, if more than one failure occurs, the lot shall be rejected. If only one defective is found, a second sample of 50 shall be tested. If more than one defective is found in the second sample, the lot shall be rejected.

4.7.1.3.3 Rejected lots. If an inspection lot is rejected, disposition of the lot shall be as directed by the contracting activity. In the event the contractor is directed to rework the lot, a written procedure for the rework shall be submitted to the responsible Government engineering activity for approval before commencing the rework. Such lots shall be separate from new lots and shall be clearly identified as reworked lots in accordance with MIL-STD-1168.

4.7.1.3.4 Primer sensitivity sample. The sample for primer sensitivity shall be in accordance with table VI.

TABLE VI. *Primer sensitivity sample.*

Lot size	Primer sensitivity sample size
3-40	5
41-110	10
111-800	35
over 800	50

4.7.1.4 Group D inspection. Group D inspection shall consist of the tests specified in table VII and shall be made on sample units which have undergone and passed Group A inspection. The sample shall consist of 30 cutter assemblies randomly selected from each lot unless otherwise specified by the contracting activity (see 6.9.5).

TABLE VII. *Group D inspection.*

Inspection	Requirement paragraph	Test method paragraph	Quantity tested	Accept/reject criteria
Vibration	3.12	4.8.14	12	4.7.1.4.1
Release and non-release for firing pin	3.14.1	4.8.16	30	4.7.1.4.1
Functional test over temperature range	3.14.4	4.8.17	21	4.7.1.4.1
Functional test at high setback	3.14.5	4.8.18	9	4.7.1.4.1

4.7.1.4.1 Test failure. The cutter shall be classified as defective if it releases the firing pin under minimum force, fails to fire under maximum force, fails to meet the specified time requirement, or fails to completely sever both reefing lines. Failure of any unit to comply with requirements of 3.12 and 3.14 shall be cause for lot rejection.

4.7.1.4.2 Retest. There shall be no retests of a failed lot unless directed by the contracting activity. The retest shall be done under direction of the responsible Government engineering activity.

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4.8 Tests and examinations.

4.8.1 Visual and dimensional examinations. Visual examinations shall be made for such defects as defective metal, burr, sharp corners, foreign matter, defective protective finish, marking missing, incorrect, or illegible, thread damaged or not full, holes, slots, radii or chamfers missing. Four power (4×) magnification may be used in performing visual examination for such defects. An appropriate gage or measuring device shall be used to check all other characteristics shown on the drawing. Inspection for rubber O-rings shall be in accordance with MIL-STD-413 and MIL-A-2550.

4.8.2 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine the hardness values specified on the applicable drawing.

4.8.3 Protective finish. Test procedures for protective finish shall be in accordance with the component drawing.

4.8.4 Primer x-ray (C1). Prior to assembly into the primer retainer, the primer shall be x-rayed with the side of cup perpendicular to the rays. The x-ray negative shall be identified by item nomenclature and lot number. After screening and indicating defective primers, the x-ray photographs shall be forwarded to the responsible Government engineering activity (see 6.3).

4.8.5 Primer sensitivity. Tests shall be made on primers assembled into primer retainers. The primer shall be crimped and sealed in the retainer in accordance with the assembly drawing. Drop test at 13 inches (adjusted) using a stainless steel ball weighing 1.94 ± 0.2 ounces in a primer sensitivity tester with a primer retainer holder approved by the contracting activity. The drop height of 13 inches shall be adjusted taking into consideration the actual weight of the ball so that an energy level of 26 inch-ounces shall be achieved. All height measurements shall be made between the head of the firing pin and the bottom of the steel ball. The primer sensitivity test equipment shall be checked to insure a center impact of the steel ball on the head of the firing pin.

4.8.6 Spring load. The spring shall be assembled in the test equipment and force applied until the spring is compressed to the larger height specified in Drawing 8875988. The applied force shall be recorded. Force shall be re-applied until the unit is compressed to the smaller height specified in Drawing 8875988 and the applied force recorded. Any spring exhibiting a compression force at either height outside the load range permitted by Drawing 8875988 shall be classified defective.

4.8.7 Spring embrittlement. The spring shall be held at each end and stretched until the minimum point of permanent distortion is reached. If the spring breaks before reaching that point, it shall be classified defective. Springs so tested shall be discarded.

4.8.8 Corrosion resistance. Corrosion resistance testing shall be performed as specified in MIL-STD-810.

4.8.9 Airtightness. Each knife delay assembly shall be tested in a dry gas leak tester.

4.8.10 Waterproofness of knife assembly. The knife delay assembly shall be placed under water in a vacuum chamber. The water shall contain the wetting agent Maccon NR (sodium alkyl aryl sulfonate) or equal. The chamber shall be evacuated to a minimum pressure differential of 5 psi. Any assembly emitting bubbles, at any frequency, from one or more points shall be classified as defective and discarded. After subjection to test, the assembly shall be rinsed thoroughly in fresh running water to remove the wetting agent and dried at room temperature.

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4.8.11 Lanyard assembly pull test. The lanyard assembly shall be placed in the applicable test equipment wherein it shall be held at the firing pin release and through the lanyard eye. The specified load shall be applied. No dwell time is required. Any unit which separates shall be classified defective.

4.8.12 Radiographic examination. Each assembled cutter shall be radiographically examined in accordance with MIL-STD-453 with the exception that a fully assembled cutter shall be substituted for the penetrameter. The cutters shall be positioned for the most revealing exposure with the long axis of the cutter perpendicular to the rays of the x-ray machine. All assemblies shall be examined by radiographic plates and any observable imperfections in components or assembly shall be cause for cutter rejection. All assemblies shall be identified with serial numbers prior to examination. The cutter serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The assemblies for radiographic examination shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the assemblies displayed thereon. Any missing serial numbers are to be identified on the x-ray. The radiographic plate identification shall include the complete lot number, as stamped on the cutters, and the span of serial numbers displayed. Radiographic plates of the entire production lot shall be forwarded to the activity conducting the ballistic acceptance testing (see 6.3). Film interpretation of the radiographic plates shall verify that the final assembly is properly assembled and includes all components. Defective assemblies found during x-ray review are to be marked on the x-ray prior to shipment and those defective items shall be removed from the production lot.

4.8.13 Staking (M102). The assembled cutter shall be placed in a holding device in which the prescribed amount of torque can be applied to the retainer to verify that the staking between case and retainer meets the requirements of 3.11.

4.8.14 Vibration. The selected sample shall undergo transportation vibration testing as specified by procedure II in Test 401 of MIL-STD-331.

4.8.15 Withdrawal and non-withdrawal of cotter pin. The assembled cutter shall be firmly mounted in a holding fixture. A force equal to the minimum specified withdrawal force shall be applied, without impact, to the cotter pin so as to tend to remove the pin from the assembly. The force shall act in a direction parallel to the length of the pin and shall be applied at the eye. If the safety cotter pin is not withdrawn when the minimum force is applied, the maximum force shall be applied in the same manner as specified above. Any cotter pin which is withdrawn when the minimum force is applied or is not withdrawn when the maximum force is applied shall cause the assembly to be classified as defective.

4.8.16 Firing pin release and non-release. The firing pin assembly (Drawing 6261286) shall be secured in a fixture. A pull force shall be applied to the lanyard in the longitudinal direction in conjunction with the fixture which registers the amount of pull. If the force to release the firing pin is less than the specified minimum, the assembly shall be classified as defective.

4.8.16.1 Firing pin release in functional test. In conjunction with functioning over temperature range tests, the force required to release the firing pin shall be recorded on the firing record.

4.8.17 Functioning over temperature range. The cutter shall be mounted in the test fixture, Drawing 8873322. Two reefing lines (1000 pound tensile strength, 5/8 inch, MIL-W-5625) shall be passed through the hole in the cutter. One end of the reefing lines shall be anchored and a 50-pound load attached to the free ends of the reefing lines. The lanyard shall then be pulled to release the firing pin. A ballistic trace which records the force applied to the lanyard to release the firing pin and the delay time (see 6.9.2) shall be made for each firing.

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4.8.17.1 Temperatures. The sample for functioning over the temperature range shall be divided into three equal groups and temperature conditioned as shown in table VIII. The units shall be tested within 10 minutes after removal from the conditioning chamber. When this time is exceeded, the sample unit shall be reconditioned for a minimum of 1 hour.

TABLE VIII. *Conditioning requirements for functioning test.*

Temperature (°F)	Vibrated	Non-vibrated	High setback
First article inspection sample			
−65 to −70	4	3	
65 to 70	4	3	7
160 to 165	4	3	
Group D sample			
−65 to −70	4	3	
65 to 70	4	3	9
160 to 165	4	3	

4.8.17.2 Temperature conditioning. Each cutter shall be conditioned for a period of time not less than 4 hours nor greater than 24 hours. When reconditioning cutters at −65 °F, all condensation shall be removed from the exterior surfaces before returning the cutter to the conditioning chamber.

4.8.18 Functioning at high setback. The cutter assembled to the test fixture (Drawing 8873322) shall be initiated by release of the firing pin. Midway through the delay firing time the specified shock shall be applied in the direction of the burning of the delay. Observation shall be made for complete functioning as evidenced by complete severing of the reefing lines. This test is performed at ambient temperature.

4.9 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be level A or B as specified by the contracting activity:

Level A. In accordance with Drawing 6261285

Level B. In accordance with Drawing 6261284

5.2 Marking (M103). Marking shall be in accordance with MIL-STD-129 and the requirement of Drawings 6261284 and 6261285.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The cutter is used to cut nylon reefing line in cargo air drop systems.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and the specific issue of individual documents referenced (see 2.1)

c. Whether first article sample is required, and if so specify the test activity (see 3.2 and 4.6).

d. Production lot size and test activity (see 4.1 and 4.7).

e. Ballistic test sample if other than 30 units (see 4.7.1.4)

f. Direction for shipping radiographs of primers before assembly and of the entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.8.4, 4.8.12 and 6.3).

g. Level of marking and packing (see 5.1).

h. Items of data required for materials inspection and production lot (see 6.3).

i. Inspection sampling plan and acceptance/rejection criteria (see 4.7.1.3.1 and 6.5).

j. Safety precautions (see 6.6)

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirements for a DD Form 1423.

Ref. Paragraph	DID Number	DID Title	Tailoring
4.3	DI-MISC-80678	Certification Data Reports	None
4.8.4, 4.8.12	DI-MISC-80653	Test Reports	None
6.3.1	DI-MISC-80043	Ammunition Data Card	None

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Data cards. The contract should state the format, quantity and destination of the Ammunition Data Cards.

6.4 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample (see 3.1) and the number of items to be tested as specified in 4.6.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government,

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and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Inspection sampling plans and acceptance or rejection criteria. The sampling plans and acceptance or rejection criteria of tables IX and X have been used successfully in procuring, examining, and testing the MLU-58/B Delay Cutter Assembly. For an individual defect (single characteristic), the criteria in table IX have been used. For total defects within a major or minor group the criteria in table X have been used. The contracting activity is cautioned that any deviation from the values in tables IX and X should be warranted and verified statistically considering the criticality of the delay cutter function.

6.6 Safety precautions. The safety precaution requirements of "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as a part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosive Ashore," OP 5, are applicable.

6.7 Explosive hazard classification information.

Hazard Classification Division: 1.4
Storage Compatibility Group: S
DOT Class: C
DOT Marking - Explosive Cable Cutter

TABLE IX. *Sampling plans and acceptance criteria.*

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1-280	50[1]	0	1	2	3
281-1,200	80	1	2	3	4
1,201-3,200	125	2	3	5	6
3,201-10,000	200	3	4	7	8

[1]If sample size equals or exceeds lot size do 100 percent inspection. When 100 percent inspection is done, all nonconforming material is rejected, and all conforming material is accepted.

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TABLE X. *Total allowable defects per sample size.*

Number of characteristics in class	Sample size				
	50	80	125	200	315
		Major defects			
2-3	—	1	2	3	4
4-7	1	2	3	4	5
8-19	2	3	4	5	7
20-43	3	4	5	8	10
Over 43	4	6	8	11	14
		Minor defects			
2-3	1	2	3	4	5
4-7	2	3	4	5	7
8-16	3	4	5	8	10
17-48	4	6	8	11	14
over 48	6	8	11	17	20

6.8 Hazard notice. The cutter described herein and certain of its components are flammable, explosive, or both, and present hazards in manufacturer, handling, storage, and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.9 Definitions. For the purpose of this specification, the following definitions apply:

6.9.1 Mechanical failure. A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.9.2 Delay time. Delay time is defined as the time elapsed from firing pin release on the ballistic record until the severance of the reefing lines.

6.9.3 Defect. A defect is any non-conformance of a characteristic with specified requirements.

6.9.4 Group A inspection. Nondestructive inspection of all items acquired or all samples from an inspection lot to demonstrate product compliance with contractural requirements. Group A inspections examine characteristics most affected by variations in production processes or skills and functions vital to successful completion of design mission.

6.9.5 Group D inspection. Destructive tests that consume all of the design service life. Articles tested in Group D inspections are considered completely expended and will not be issued for service use. Tests are performed on few samples based on production quantities.

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6.10 Subject terms (key word) listing:

Explosive cable cutter
Parachute reefing line cutter
Reefing line cutter

Custodian
Navy-OS
Air Force-99

Review Activity
Air Force-70

Preparing Activity
Navy-OS

(Project No. 1377-966)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-82762	CUTTER, PROPELLANT ACTUATED, MLU-58/B

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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